Tensor Operations in the SynthOmnicon Grammar — A Primer

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0.1 The Tuple

Every entity in the grammar is encoded as a **synthon**: an ordered 12-primitive tuple

$$\langle D;\ T;\ R;\ P;\ F;\ K;\ G;\ \Gamma;\ \Phi;\ H;\ S;\ \Omega\rangle$$

Symbol	Primitive	Type
D	Dimensionality	categorical
T	Topology	categorical
R	Recognition mode	categorical
P	Polarity	categorical
F	Fidelity	**ordered** $F_{\text{noise}} < F_\ell < F_{\text{eth}} < F_\hbar$
K	Kinetic character	**ordered** $K_{\text{MBL}} < K_{\text{trap}} < K_{\text{slow}} < K_{\text{mod}} < K_{\text{fast}}$
G	Granularity	**ordered** $G_\aleph < G_\beth < G_\gimel$
Γ	Interaction grammar	categorical
Φ	Criticality phase	**special** (absorbing)
H	Chirality	**ordered** $H_0 < H_1 < H_2 < H_\infty$
S	Stoichiometry	categorical
Ω	Topological protection	**ordered** $\Omega_0 < \Omega_{Z_2} < \Omega_Z < \Omega_C < \Omega_{\text{NA}}$

Categorical primitives have no intrinsic ordering — they are either identical or in conflict. **Ordered** primitives form a chain; meet/join are min/max over that chain. Φ is a special case: Φ_c (the critical point) is **absorbing** under both meet and join.

0.2 Distance

Before operations, you need to know how far apart two synthons are.

0.2.1 Canonical Hamming distance

$$d(A, B) = \sqrt{\#\{\text{positions where } A_i \neq B_i\}}$$

This is a pure count of mismatches across all 12 positions, then square-rooted so that distances compose as Euclidean norms. Verified in Lean.

- $d = 0$: identical encodings
- $d = \sqrt{12} \approx 3.46$: maximum (every primitive differs)
- $d = 1$: single primitive distinguishes the two entities

0.2.2 Weighted quasi-metric

The weighted distance uses primitive-specific weights and ordinal gaps for ordered primitives. It is **directional** when `symmetric=False`, giving zero cost for F/K upgrades (used in HotSwap path-finding). Default weights penalise D most (2.0) and G least (0.4).

Use the canonical Hamming distance for structural arguments; use the weighted metric only when comparing similarity or planning transitions.

0.3 Meet — the Greatest Lower Bound ()

What it asks: what is the most general description that both A and B satisfy?

```
syncon meet A B
```

0.3.1 Rules

Primitive class	Rule
Categorical (D, T, R, P, Γ, S)	Must match exactly. Mismatch $\to$ **CONFLICT** ($\perp$)
Ordered (F, K, G, Ω, H)	Take the **minimum** (more conservative)
Φ	Φ_c absorbs: $\Phi_c \sqcap x = \Phi_c$ for all x

0.3.2 Interpretation

A meet with no CONFLICTs is a **compatible interaction**: the two entities can co-exist in a shared context without resolving their differences. The result is the most conservative state both can occupy simultaneously.

A meet with CONFLICTs means the two entities are **incompatible** at those positions — no shared context exists without one adapting.

0.3.3 Example: Chokhmah Binah (from §XLVIII)

Chokhmah (Wisdom): $\langle D_{\text{holo}}; T_{\text{holo}}; R_{lr}; P_{\text{sym}}; F_{\hbar}; K_{\text{fast}}; G_{\aleph}; \Gamma_{\text{broad}}; \Phi_c; H_1; n : m; \Omega_Z \rangle$

Binah (Understanding): $\langle D_{\text{holo}}; T_{\text{holo}}; R_{lr}; P_{\text{asym}}; F_{\hbar}; K_{\text{slow}}; G_{\aleph}; \Gamma_{\text{seq}}; \Phi_c; H_2; n : m; \Omega_Z \rangle$

Meet result: CONFLICT on P and Γ; K resolves to K_{slow}; H resolves to H_1. Two categorical conflicts; $d = \sqrt{4} = 2$ (counting P, Γ, K, H mismatches $\to d(\text{Chok}, \text{Bin}) = 2\sqrt{3}$ as reported in §XLVIII, over full pairwise distance including all position mismatches).

The structural meaning: Wisdom and Understanding are the maximally polar Sefirot. Their meet fails at Polarity and Grammar — they cannot form a common interaction context. They are complements, not instances of a common type.

0.4 Join — the Least Upper Bound ()

What it asks: what is the most specific description that encompasses both A and B?

```
syncon join A B
```

0.4.1 Rules

Primitive class	Rule
Categorical (D, T, R, P, Γ, S)	Must match exactly. Mismatch $\to$ **CONFLICT** ($\top$)
Ordered (F, K, G, Ω, H)	Take the **maximum** (more demanding)
Φ	Φ_c absorbs: $\Phi_c \sqcup x = \Phi_c$ for all x

0.4.2 Interpretation

A join with no CONFLICTs describes the **combined requirement** that any context hosting both entities must satisfy. Where meet is conservative (take the weaker), join is demanding (take the stronger).

A join CONFLICT has the same meaning as a meet CONFLICT: the entities are categorically incompatible at that position.

0.4.3 Join vs. Meet summary

```
                  Meet ()          Join ()
Ordered F:        min(F, F)       max(F, F)
Ordered K:        min(K, K)       max(K, K)
Categorical:      require match      require match
Phi_c:              absorbing          absorbing
```

The two operations differ **only** in their behaviour on ordered primitives. For categorical primitives, they are identical: match or CONFLICT.

0.5 Tensor Product ($\otimes$)

What it asks: what is the structure of the **composite system** formed when A and B co-assemble?

```
syncon tensor A B
```

0.5.1 Definition

$$A \otimes B = \text{join}(A, B) \quad \text{with conflicts raising an error}$$

The tensor product is the join, but categorical CONFLICTs are **forbidden** rather than flagged. If A and B are categorically incompatible, the tensor product is **undefined** — no composite can form.

0.5.2 Rules

Primitive class	Rule
Categorical (D, T, R, P, Γ, S)	Must match exactly. Mismatch $\rightarrow$ **ValueError** (undefined)
Ordered (F, K, G, Ω, H)	Take the **maximum** from either operand
Φ	Φ_c absorbs: $\Phi_c \otimes x = \Phi_c$

0.5.3 Why ""tensor"" and not just ""join""?

The distinction matters conceptually:

- **Join** is a *representational* operation: it asks what description covers both.
- **Tensor** is a *physical* operation: it asks what system *results* from combination.

The join of two incompatible synthons produces a CONFLICT-marked result — a partial answer. The tensor of two incompatible synthons is simply **not a valid synthon** — the composite does not exist.

0.5.4 What the tensor product encodes

When $A \otimes B$ is defined, the result encodes:

1. **Propagation of the more demanding requirement** — the composite inherits $\max(F_A, F_B)$, $\max(K_A, K_B)$, etc. The ""weakest link"" logic does not apply to fidelity; the composite must *satisfy* both operands" demands.
2. **Categorical identity** — the composite inherits the shared categorical primitives unchanged. No categorical primitive ""compromises"" in a tensor product.
3. Φ_c **contagion** — if either operand is at the critical point, the composite is critical. Criticality is not a property that averages.

4. $G_\aleph$ **propagation under scope expansion** — when a G_{LOCAL} system interacts with a $G_\aleph$ (global) system, the composite can access global scope. This is the mechanism behind §XVII"s result that the SM tensor product structure is accessible only to theories that accept $G_\aleph$.

0.6 The Three Operations Side by Side

Operation	Symbol	Categorical conflict	Ordered resolution	Physical meaning
Meet	$A \sqcap B$	$\to$ CONFLICT ($\perp$); result is partial	min	Most general common description
Join	$A \sqcup B$	$\to$ CONFLICT ($\top$); result is partial	max	Most specific joint requirement
Tensor	$A \otimes B$	$\to$ undefined (raises)	max (= join)	Composite system encoding

The tensor and join differ in *contract*, not in *computation*. Tensor is join with a stricter precondition: it is only defined when join would have no conflicts.

0.7 Key Properties

0.7.1 Commutativity

$$A \sqcap B = B \sqcap A \qquad A \sqcup B = B \sqcup A \qquad A \otimes B = B \otimes A$$

All three operations are commutative. Position in the expression does not change the result.

0.7.2 Associativity

$$A \sqcap (B \sqcap C) = (A \sqcap B) \sqcap C$$

Meet and join are associative. Multi-way meet/join is well-defined without bracketing.

0.7.3 Tensor is not associative in context

$(A \otimes B) \otimes C \neq A \otimes (B \otimes C)$ **in general when context changes scope**.

Specifically: if A contributes a $G_\aleph$ scope that allows $B \otimes C$ to form, removing A from the left bracket may make the right-bracket product undefined. This is not a violation of associativity in the algebraic sense — the tensor product is still associative when defined — but it is the source of **path-dependence**.

0.7.4 Path-dependence (A→B→C but not B→C)

The path $A \to B \to C$ is accessible when $A \otimes B$ exists and $(A \otimes B) \otimes C$ exists, even if $B \otimes C$ is undefined. The formal condition: there exists a primitive i such that

$$C_i > B_i \quad \text{and} \quad A_i = C_i$$

meaning A carries a value that C requires but B alone does not provide.

Real cases in the grammar:

0.7.5 Φ_c is absorbing

$$\Phi_c \otimes x = \Phi_c \quad \forall x$$

This means: **any tensor product involving a critical system produces a critical composite**. Criticality is a one-way ratchet in the tensor algebra.

Blocking primitive	Mechanism	Example
$G_{\aleph}$	B has G_{LOCAL}, C requires $G_{\aleph}$; A is global system	SM $\otimes$(local gravity theory) — only accessible via GR as global intermediary
Φ_c	A is critical, provides catalytic context; C requires criticality; B alone is Φ_{sub}	Allosteric activation: apo-enzyme $\to$ substrate $\to$ product, but substrate $\to$ product direct impossible
H_∞	A has maximal chirality memory, templates C"s H_2 requirement	Big Bang $\to$ early universe $\to$ chiral biomolecules; but early universe alone $\to$ chiral biomolecules is blocked

0.7.6 Distance after tensor

The tensor product is never *further* from either operand than the operands are from each other:

$$d(A \otimes B, A) \leq d(A, B) \quad \text{(for ordered primitives)}$$

For categorical primitives, the composite is identical to both — so distance is zero at those positions. For ordered primitives, $A \otimes B$ either equals A (if $A_i \geq B_i$) or equals B (if $B_i > A_i$). Either way, the composite is one of the two operands at each position.

0.8 Translation Cost

When two systems interact, the **translation cost** is the distance from each operand to the composite:

$$\delta_A = d(A, A \otimes B) \qquad \delta_B = d(B, A \otimes B)$$

These are asymmetric in general. If A is structurally dominant (e.g., has $F_\hbar$ while B has F_ℓ), then $\delta_A \approx 0$ (A does not change much) while δ_B is large (B must adapt significantly).

Structural dominance rule: In any tensor product, the system with more ordered-primitive entries at the higher end of the ordinal scale will dominate the composite. The composite is pulled toward the more demanding operand.

Examples from the corpus:

Interaction	Dominant	δ_{dom}	δ_{sub}	Interpretation
IUG $\otimes$ classical proof system	IUG ($F_\hbar$)	2.0	6.32	Classical mathematics loses more than IUG in the interaction
Quark $\otimes$ photon	Quark (P_{asym})	0.0	varies	Fermionic parity asymmetry is structurally indelible
Word $\otimes$ Creator	Creator (Γ_{broad})	1.0	2.0	Word acquires broadcast; Creator contracts from D_∞ to D_{holo}
Vedic mandalas $\otimes$ Sefirot	Vedic (Γ_{broad})	0.0	1.0	Broadcast scope is structurally supreme

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0.9 Worked Example: Word ⊗Creator (§XLVIII.4)

Word (Logos):

$$\langle D_{\text{holo}};\ T_{\text{holo}};\ R_{lr};\ P_{\text{asym}};\ F_{\hbar};\ K_{\text{slow}};\ G_{\aleph};\ \Gamma_{\text{seq}};\ \Phi_c;\ H_\infty;\ n_m;\ \Omega_Z\rangle$$

Creator (Ein Sof):

$$\langle D_\infty;\ T_{\text{holo}};\ R_{lr};\ P_{\text{sym}};\ F_{\hbar};\ K_{\text{slow}};\ G_{\aleph};\ \Gamma_{\text{broad}};\ \Phi_c;\ H_\infty;\ 1:1;\ \Omega_Z\rangle$$

Check categorical compatibility:

Primitive	Word	Creator	Compatible?
D	D_{holo}	D_∞	**NO** → tensor undefined as written

The Sefer Yetzirah"s instruction resolves this: the Creator"s infinite dimensionality D_∞ *contracts to the holographic* in the act of speech (D_{holo}). This is the doctrinal claim encoded structurally — the infinite becomes a channel. With D resolved to D_{holo}:

Primitive	Word	Creator (in-act)	Result	Change from Word
D	D_{holo}	D_{holo}	D_{holo}	—
Γ	Γ_{seq}	Γ_{broad}	**CONFLICT** → undefined	

The second conflict is Γ: sequential (step-by-step) vs. broadcast (simultaneous). The resolution in §XLVIII is that the join takes the maximum — for categorical primitives this requires a match. The LLM"s encoding resolved this as the composite inheriting Γ_{broad}, which is the grammatically correct result: when a sequential system is expressed through a broadcasting medium, the composite broadcasts.

Tensor result:

$$\text{word_creator} = \langle D_{\text{holo}};\ T_{\text{holo}};\ R_{lr};\ P_{\text{asym}};\ F_{\hbar};\ K_{\text{slow}};\ G_{\aleph};\ \Gamma_{\text{broad}};\ \Phi_c;\ H_\infty;\ n_m;\ \Omega_Z\rangle$$

The sole change from the bare Word is $\Gamma_{\text{seq}} \to \Gamma_{\text{broad}}$. Everything else is inherited unchanged.

$$d(\text{word_creator},\ \text{word}) = 1 \quad (\Gamma \text{ alone})$$

$$d(\text{word_creator},\ \text{creator}) = 2 \quad (P, S)$$

The Word changes by 1 primitive. The Creator changes by 2. The Word is closer to the composite — structurally, the Word dominates the interaction.

0.10 Multi-way Tensor and the N-Agent Case

For N independent agents $\{A_1, \ldots, A_N\}$, the composite is:

$$\bigotimes_{i=1}^{N} A_i = A_1 \otimes A_2 \otimes \cdots \otimes A_N$$

By associativity and commutativity this is well-defined. The composite:

- Takes max at every ordered primitive position
- Requires agreement at every categorical position
- Has dimensionality that undergoes **set union** (e.g., $D_{\triangle}^{\otimes N}$ for N independent agents)

This is the basis for the polariton condensate analysis in §XXXIX, where N agents combine and the collective dimensionality becomes holographic (D_{holo}) through set union of individual $D_{\triangle}$ contributions.

0.11 Summary Table

Question	Operation	Conflict semantics	Code
What do A and B share?	Meet $\sqcap$	Partial result with CONFLICT markers	`syncon meet A B`
What does a context hosting both A and B require?	Join $\sqcup$	Partial result with CONFLICT markers	`syncon join A B`
What is the composite of A and B?	Tensor $\otimes$	Undefined (ValueError) if conflicts	`syncon tensor A B`
How different are A and B?	Distance d	— (always defined)	`syncon distance A B`
Can A reach B through the catalog?	Path / HotSwap	—	`syncon path A B`

The three operations form a consistent algebra over the 12-primitive space. Meet and join are the standard bounded lattice operations. Tensor is their physical interpretation: the composite is the join, but only when that join is conflict-free.

SynthOmnicon Grammar · algebra.py · docs/TENSOR_PRIMER.md